Configuration Summary

TPLD2001DGSR

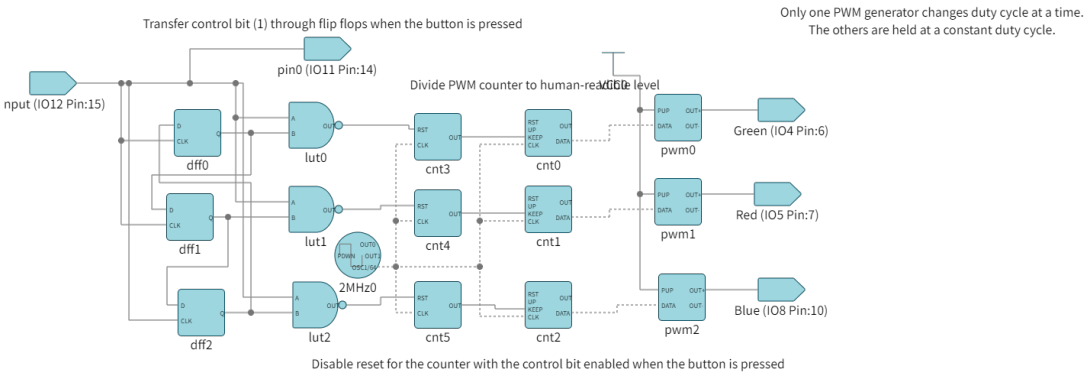
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Summary

This document is to provide information related to the design. The information available in this document is an image of the schematic, an estimation of power consumed, an input and output table, a set of tables indicating how the blocks are configured, frequencies propogating through the device, and the hex code that was pushed into registers. For general device information such as electrical characteristics refer to the datasheet.

Schematic Diagram



TPLD2001 INPUT PIN TABLE

Name	Pin	Schematic Name	In Mode	Pull	Resistor
IN0	2	Unused		Pull Down	1M Ohms
IO1	3	Unused		Pull Down	1M Ohms
IO2	4	Unused		Pull Down	1M Ohms
IO3	5	Unused		Pull Down	1M Ohms
IO6	8	Unused		Pull Down	1M Ohms
IO7	9	Unused		Pull Down	1M Ohms
IO9	12	Unused		Pull Down	1M Ohms
IO10	13	Unused		Pull Down	1M Ohms
IO12	15	Input	Digital Input without Schmitt trigger	Pull Down	Floating
IO13	16	Unused		Pull Down	1M Ohms
IO14	17	Unused		Pull Down	1M Ohms
IO15	18	Unused		Pull Down	1M Ohms
IO16	19	Unused		Pull Down	1M Ohms
IO17	20	Unused		Pull Down	1M Ohms

TPLD2001 OUTPUT PIN TABLE

Name	Pin	Schematic Name	Out Mode	Strength
IO4	6	Green	Push-Pull 1X	
IO5	7	Red	Push-Pull 1X	
IO8	10	Blue	Push-Pull 1X	
IO11	14	pin0	Push-Pull 1X	

/ti/tpld/SYSTEM is allocated to MISC

Configurable	Description
Register Read Lock	false
Register Write Lock	false
Program ID	0
GPIO Quick Charge Enable	false
OTP Write Lock	false
Prebias Control	UNEXPECTED
Read Data Sync Control	false
User Read/Write Lock	Read/Write to all non-reserved, non-read-only registers
Bandgap Control	Auto

2MHz0 is allocated to OSC1

Configurable	Description
PDWN	Internal GND
Power Control Source Select	From register
cntl_in	Power down
Power Mode	Auto Power On
OUT0 Second Stage Divider	/1
OUT1 Second Stage Divider	/1
Clock Pre Divider	/1
Clock Source	Internal RC Oscillator

Blue is allocated to IO8

Configurable	Description
IN	pwm2.OUT0
in_ctl	Digital Input without Schmitt trigger
Output Mode	Push-Pull 1X
Pull	Pull Down
Pull Up/Down Resistor Value	Floating

cnt0 is allocated to CNT6_FSM0

Configurable	Description
CLK	2MHz0.OSC1/64
IN	Internal GND
KEEP	cnt3
UP	Internal GND
Initial Value	Bypass initial
Output Polarity	Non-inverted
Control Data	255
KEEP Sync Bypass	2-DFF sync
RST Sync Bypass	2-DFF sync
UP Sync Bypass	2-DFF sync

cnt1 is allocated to CNT7_FSM1

Configurable	Description
CLK	2MHz0.OSC1/64
IN	Internal GND
KEEP	cnt4
UP	Internal GND
Initial Value	Bypass initial
Output Polarity	Non-inverted
Control Data	255
KEEP Sync Bypass	2-DFF sync
RST Sync Bypass	2-DFF sync
UP Sync Bypass	2-DFF sync

cnt2 is allocated to CNT8_FSM2

Configurable	Description
CLK	2MHz0.OSC1/64
IN	Internal GND
KEEP	cnt5
UP	Internal GND
Initial Value	Bypass initial
Output Polarity	Non-inverted
Control Data	255
KEEP Sync Bypass	2-DFF sync
RST Sync Bypass	2-DFF sync
UP Sync Bypass	2-DFF sync

cnt3 is allocated to LUT3_10

Configurable	Description
Initial Value	Bypass initial
Output Polarity	Non-inverted
Control Data	254
RST Sync Bypass	2-DFF sync

cnt4 is allocated to LUT3_8

Configurable	Description
Initial Value	Bypass initial
Output Polarity	Non-inverted
Control Data	254
RST Sync Bypass	2-DFF sync

cnt5 is allocated to CNT9_FSM3

Configurable	Description
CLK	2MHz0.OSC1/64
IN	lut2
Initial Value	Bypass initial
Output Polarity	Non-inverted
Control Data	254
keep_sync	2-DFF sync
RST Sync Bypass	2-DFF sync
up_sync	2-DFF sync

dff0 is allocated to LUT2_2

Configurable	Description
CLK	Input
IN	dff2
Invert Clock Input	false
Initial Polarity	High
Mode	DFF
Generate Inverted Output	false

dff1 is allocated to LUT2_1

Configurable	Description
CLK	Input
IN	dff0
Invert Clock Input	false
Initial Polarity	Low
Mode	DFF
Generate Inverted Output	false

dff2 is allocated to LUT2_0

Configurable	Description
CLK	Input
IN	dff1
Invert Clock Input	false
Initial Polarity	Low
Mode	DFF
Generate Inverted Output	false

Green is allocated to IO4

Configurable	Description
IN	pwm0.OUT0
in_ctl	Digital Input without Schmitt trigger
Output Mode	Push-Pull 1X
Pull	Pull Down
Pull Up/Down Resistor Value	Floating

Input is allocated to IO12

Configurable	Description
OE	GND
Input Mode	Digital Input without Schmitt trigger
out_ctl	Push-Pull 1X
Pull	Pull Down
Pull Up/Down Resistor Value	Floating

lut0 is allocated to LUT4_1

Configurable	Description
A	Input
B	dff0
cfg	NAND

lut1 is allocated to LUT4_2

Configurable	Description
A	Input
B	dff1

cfg	NAND

lut2 is allocated to LUT4_3

Configurable	Description
A	Input
B	dff2
cfg	NAND

pin0 is allocated to IO11

Configurable	Description
IN	Input
OE	VCC
in_ctl	Digital Input without Schmitt trigger
Output Mode	Push-Pull 1X
Pull	Pull Down
Pull Up/Down Resistor Value	Floating

pwm0 is allocated to PWM_GEN0

Configurable	Description
PDWN	VCC0
Source Select	CNTFSM0
Invert Output -	false
Invert Output +	false
Deadband Time	No Deadband

pwm1 is allocated to PWM_GEN1

Configurable	Description

PDWN	VCC0
Source Select	CNTFSM1
Invert Output -	false
Invert Output +	false
Deadband Time	No Deadband

pwm2 is allocated to PWM_GEN2

Configurable	Description
PDWN	VCC0
Source Select	CNTFSM2
Invert Output -	false
Invert Output +	false
Deadband Time	No Deadband

Red is allocated to IO5

Configurable	Description
IN	pwm1.OUT0
in_ctl	Digital Input without Schmitt trigger
Output Mode	Push-Pull 1X
Pull	Pull Down
Pull Up/Down Resistor Value	Floating

Frequencies

Port	Frequency

OTP Map in Intel Hex Format

```
:10020000000000000000484a00004c0000000d7f86
:1002100000000000000000000000000d140d150d90
:1002200013000000000000000000000000000000bd
:10023000000000000000000000000002500009b
:1002400000000024000000000000000d1500000d5d
:10025001400000d13000000000000000000000006c
:10026000000000000000000000000000000000090
:10027000000000000000000000000000000000215f
:10028000000001f0000004700260000007f7f7f67
:10029000000000000000000000000000000000060
:1002a000000000000000000000000000000000050
:1002b000000000000000000000000000000000040
:1002c000000000000000000000000000000000030
:1002d000000000000000000000000000000000020
:1002e000000000000000000000000000000000010
:1002f000000000000000000000000000000000000
:10030000303030308080303080303000030303060
:10031000303000000000000000000000000000080
:10032000000000007000000000040000000000c5
:10033000000000000000000000000000000000c0
:10034000000000000077777777770000000e6
:1003500000000000000f000000000f0000000fec2
:100360004f00010000f000000000000000000050
:10037000000000fe004f0001000000f00000ff4cf7
:1003800000ff4c00ff4c00fe4f0000000000240069
:10039000000000000000000000000000000000060
:1003a000000000000000000000000000000000050
:1003b000000000000000000000000000000000040
:1003c000000000000000000000000000000000030
:1003d000000000000000000000000000000000020
:1003e000000000000000000000000000000000010
:1003f00000000300000000001100000000000d7eb2a :00000001FF
```